

24-2 HSRP Troubleshooting - Answer Key

In this lab you will troubleshoot HSRP.

HSRP Troubleshooting

- 1) HSRP was previously configured on the R1 and R2 routers and everything was working fine until last night. There was an outage on R1 and all the clients lost network connectivity. R1 has been restored but management are concerned that the issue will reoccur.

Your task is to troubleshoot the configuration and ensure that all PCs maintain network connectivity when there is an outage of either router. R1 must be the preferred router when both are available.

The exercise is complete when you can shut down interface GigabitEthernet 0/1 on R1 and HSRP fails over to R2. When you bring the interface back up again HSRP must fail back to R1.

Note that some commands may be disabled during the CCNA exam so you should know the different troubleshooting commands.

You must not use the 'show running-configuration' command (or any derivative of it such as 'show run | begin interface') in this exercise.

Give HSRP enough time to fail over when testing.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

First check connectivity from the PCs to their HSRP default gateway.

```
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time=11ms TTL=255
Reply from 10.10.10.1: bytes=32 time=10ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=25ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 25ms, Average = 11ms
```

Check the HSRP configuration on R1.

```
R1#show standby
GigabitEthernet0/1 - Group 1 (version 2)
State is Active
21 state changes, last state change 14:12:11
Virtual IP address is 10.10.10.1
Active virtual MAC address is 0000.0C9F.F001
Local virtual MAC address is 0000.0C9F.F001 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 1.557 secs
Preemption enabled
Active router is local
Standby router is unknown, priority 100
Priority 110 (configured 110)
Group name is hsrp-Gig0/1-1 (default)
```

R1 is the active version 2 HSRP router for Group 1 but the standby is unknown. Check the HSRP configuration on R2.

```
R2#show standby
GigabitEthernet0/1 - Group 1
State is Active
8 state changes, last state change 14:11:59
Virtual IP address is 10.10.10.2
Active virtual MAC address is 0000.0C07.AC01
Local virtual MAC address is 0000.0C07.AC01 (v1 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 2.048 secs
Preemption enabled
Active router is local
Standby router is unknown
Priority 120 (configured 120)
Group name is hsrp-Gig0/1-1 (default)
```

R2 is also an active router, but it is using HSRP version 1 and the virtual IP address 10.10.10.2. This is wrong, the version should be 2 and PCs default gateway is 10.10.10.1. Correct this.

```
R2(config)#interface g0/1
R2(config-if)#standby 1 ip 10.10.10.1
R2(config-if)#standby version 2
```

Wait for HSRP to be negotiated between the two routers then check the status.

```
R2#show standby
GigabitEthernet0/1 - Group 1 (version 2)
State is Active
10 state changes, last state change 00:05:16
Virtual IP address is 10.10.10.1
Active virtual MAC address is 0000.0C9F.F001
Local virtual MAC address is 0000.0C9F.F001 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 2.201 secs
Preemption enabled
Active router is local
Standby router is 10.10.10.2, priority 110 (expires in 6 sec)
Priority 120 (configured 120)
Group name is hsrp-Gig0/1-1 (default)
```

HSRP is working now but R2 is the active router because it has a priority of 120 compared to R1's priority of 110. Change R2's priority to be lower than R1.

```
R2(config)#int g0/1
R2(config-if)#standby 1 priority 100
```

Verify that R1 has pre-emption configured and becomes the active router.

```
R2#show standby
GigabitEthernet0/1 - Group 1 (version 2)
State is Standby
14 state changes, last state change 00:08:37
Virtual IP address is 10.10.10.1
Active virtual MAC address is 0000.0C9F.F001
Local virtual MAC address is 0000.0C9F.F001 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.056 secs
Preemption enabled
Active router is 10.10.10.2, priority 110 (expires in 8 sec)
MAC address is 0000.0C9F.F001
Standby router is local
Priority 100 (default 100)
Group name is hsrp-Gig0/1-1 (default)
```

Shut down interface GigabitEthernet 0/1 on R1 to test HSRP.

```
R1(config)#int g0/1
R1(config-if)#shutdown
```

Verify R2 fails over to HSRP active.

```
R2#show standby
GigabitEthernet0/1 - Group 1 (version 2)
State is Active
15 state changes, last state change 00:10:50
Virtual IP address is 10.10.10.1
Active virtual MAC address is 0000.0C9F.F001
Local virtual MAC address is 0000.0C9F.F001 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.762 secs
Preemption enabled
Active router is local
Standby router is unknown, priority 110
Priority 100 (default 100)
Group name is hsrp-Gig0/1-1 (default)
```

Verify the PCs still have connectivity to their default gateway.

```
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Request timed out.
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Bring interface GigabitEthernet 0/1 back up on R1 and verify HSRP fails back.

```
R1(config)#int g0/1
R1(config-if)#no shut

R1#show standby
GigabitEthernet0/1 - Group 1 (version 2)
State is Active
27 state changes, last state change 00:16:25
Virtual IP address is 10.10.10.1
Active virtual MAC address is 0000.0C9F.F001
Local virtual MAC address is 0000.0C9F.F001 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 1.386 secs
Preemption enabled
Active router is local
Standby router is unknown, priority 100 (expires in 411903
sec)
Priority 110 (configured 110)
Group name is hsrp-Gig0/1-1 (default)
```